

Containerization and Deployment Guide

1. Base Images: Standardize on slim official Linux images (python:3.11-slim or node:20-alpine) to reduce security vulnerability surfaces.
2. Multi-Stage Builds: Leverage multi-stage Dockerfiles to keep production container images under 200MB.
3. Security Scanning: Docker images must be scanned for vulnerabilities via Trivy prior to registry push.
4. Kubernetes Deployment: Deploy services using Helm charts configured with resource limits and readiness probes.